Approximating Euler's number

1

a 54.5982 **b** 0.3679 **c** 1.2214 **d** 8.2436

e 1.4715 **f** 0.0163

2

a $2.718\,280$ **b** $2.718\,281\,828$

3

a $2.012\,171$ **b** $2.013\,753$ **c** $0.001\,582$

Exponential expressions

4

a $7e^x$ **b** $9e^x$ **c** $3e^x$ **d** e^x

e $12e^{2x} + 2e^x$ **f** $-5e^{2x} - 8e^x$ **g** e^{x+2} **h** e^{5x}

i e^{5x} **j** e^{7x} **k** $2e^x - 3$ **l** $\dfrac{e^{3x} + 7e^{2x}}{6}$

5

a $e^{2x} + 3e^x$ **b** $e^{3x} - 9e^{2x}$

c $e^{2x} + 7e^x + 10$ **d** $e^{2x} - 3e^x - 10$

e $2 + 2e^x$ **f** $3 + 3e^{2x}$

g $2e + e^{2x} + e^{-2x}$ **h** $e^{2x} - 2e^{-x} + e^{-4x}$

6

a $4\left(2e^x + 3\right)$ **b** $4e^x(2e^x + 3)$

c $4e^{3x}\left(2e^x - 3\right)$ **d** $e^{4x}(8 - e^{3x})$

e $\left(e^x - 3\right)\left(e^x - 2\right)$ **f** $(e^x + 3)(e^x + 3)$

g $\left(e^x + 4\right)\left(e^x + 2\right)$ **h** $(e^{2x} - 5)(e^{2x} - 2)$

Exponential graphs

7

a For $x > 0$, the graph of $y = e^x$ will lie above the graph of $y = (2)^x$ and below the graph of $y = (3)^x$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

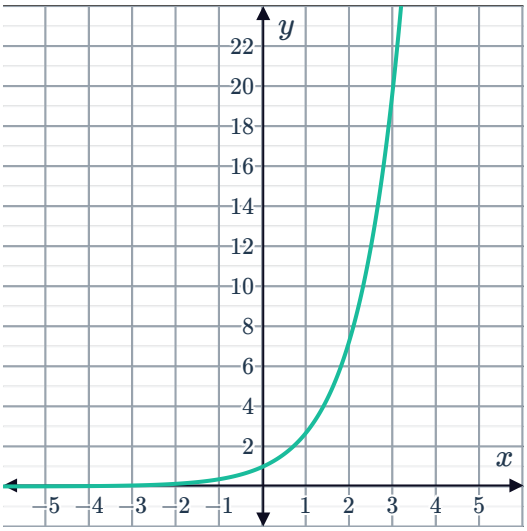
b For $x < 0$, the graph of $y = e^x$ will lie above the graph of $y = (3)^x$ and below the graph of $y = (2)^x$.

8

a

x	-3	-2	-1	0	1	2	3
$f(x)$	0.05	0.14	0.37	1.00	2.72	7.39	20.09

b



9

a No

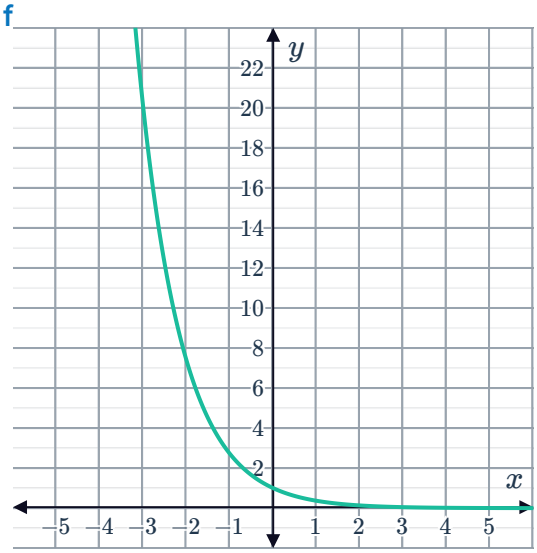
b
i 0 **ii** ∞

c No

d 1

e 0

f



10 $(0, 1)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11

a

x	$f(x)$
-5	-0.007
-4	-0.018
-3	-0.050
-2	-0.135
-1	-0.368
0	-1
1	-2.718
2	-7.389
3	-20.086
4	-54.598
5	-148.413
10	$-22\,026.5$

b No

c No

d Decreasing

e As x increases, the function decreases at a faster and faster rate.

Transformations of exponential graphs

12

a **i** No **ii** No **iii** No

b **i** No **ii** Yes **iii** Yes

c **i** Yes **ii** Yes **iii** Yes

d **i** No **ii** Yes **iii** Yes

13

a Increasing **b** Decreasing **c** Decreasing **d** Increasing

14

a A dilation by a factor of $\frac{1}{4}$ from the y-axis and a vertical translation of 1 unit upwards.

b A dilation by a factor of 5 from the x-axis and a horizontal translation of 4 units to the right.

c A dilation by a factor of 5 from the x-axis and a reflection about the y-axis.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

15

a $y = e^{x+4} + 3$ **b** $y = e^{x+2} - 5$ **c** $y = 2e^{x} - 5$ **d** $y = -e^{x-2} + 3$

e $y = e^{\frac{x}{13}} - 2$ **f** $y = e^{3x} - 2$

16

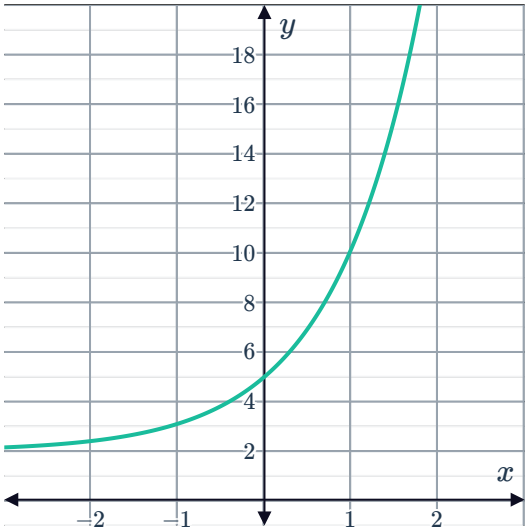
a

i $y = 3e^{x} + 2$

ii $y = 2$

iii 5

iv



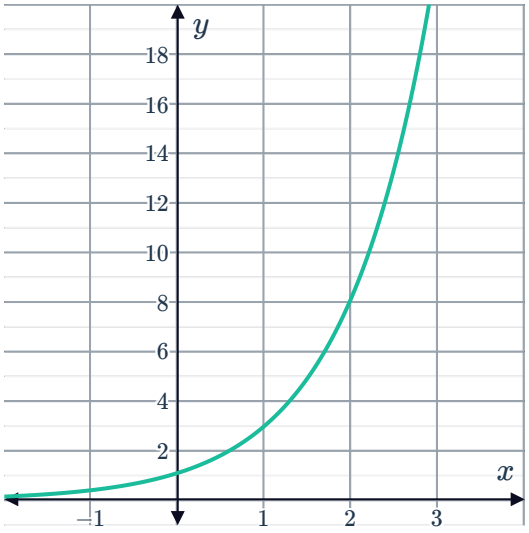
b

i $y = 3e^{x-1}$

ii $y = 0$

iii $\frac{3}{e}$

iv



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

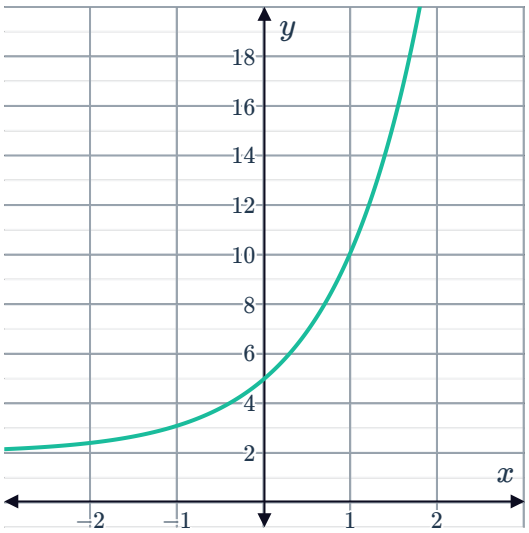
c

i $y = 3e^x + 2$

ii $y = 2$

iii 5

iv



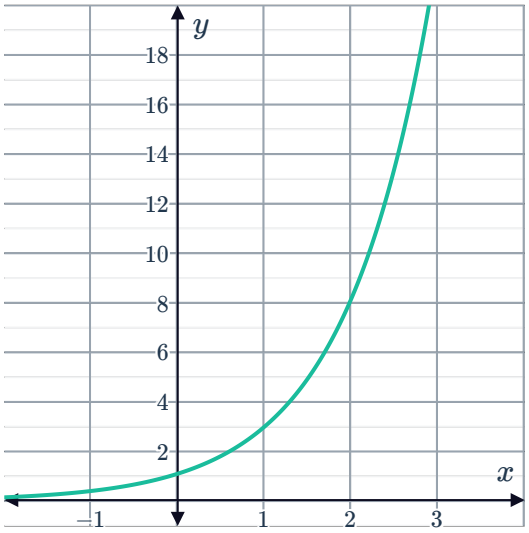
d

i $y = 3e^{x-1}$

ii $y = 0$

iii $\frac{3}{e}$

iv



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

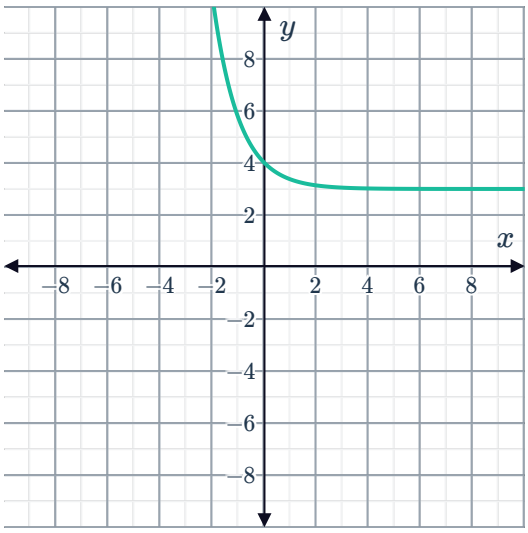
e

i $y = e^{-x} + 3$

ii $y = 3$

iii 4

iv



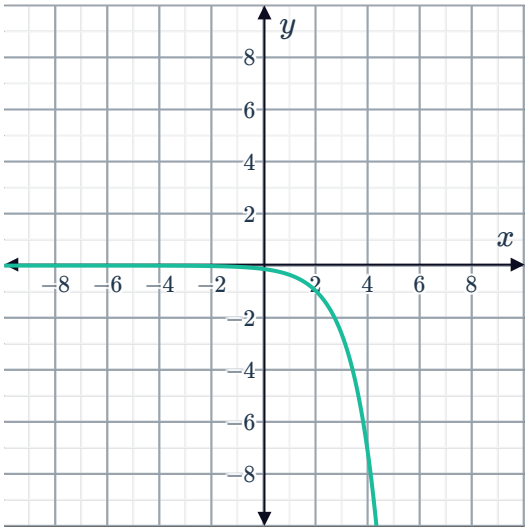
f

i $y = -e^{x-2}$

ii $y = 0$

iii $y = -\dfrac{1}{e^2}$

iv



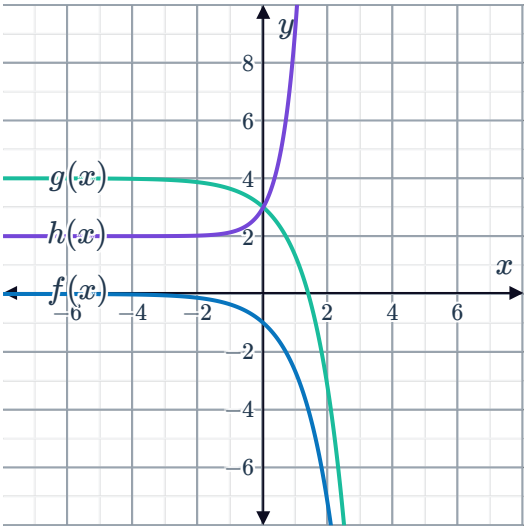
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

17

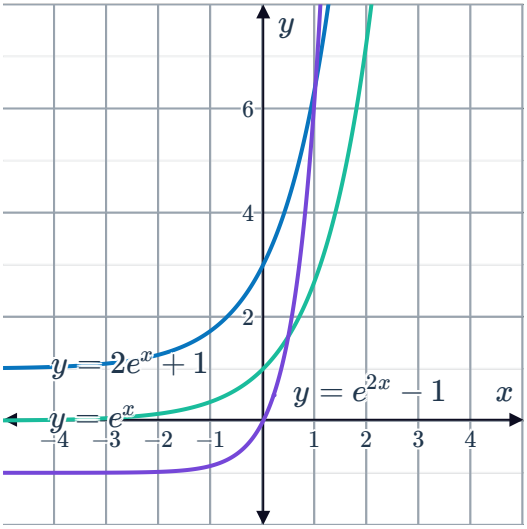
a

i $y = 0$ ii $y = 4$ iii $y = 2$

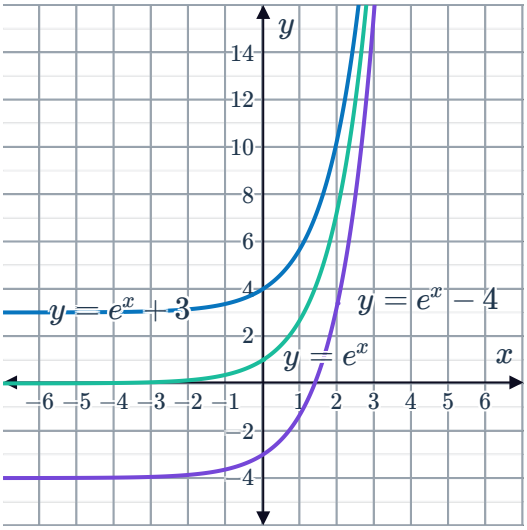
b



18

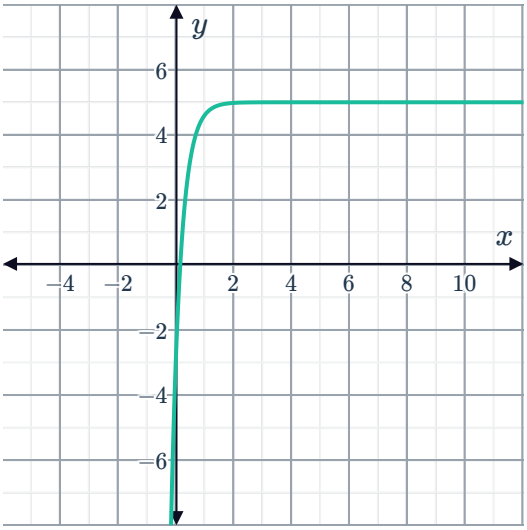


19



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

20



© ClipSAT: By Mr. Mohamed Abdallah +966597688647